

ENGM 204
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Blank Page™

The Journey Is Yours



Blank Page™ combines empathetic AI and intuitive UX to help students express, reflect, and heal

Endeavor AI is a purpose-driven startup tackling the global student mental health crisis through empathetic AI, beginning with its flagship product, Blank Page™



- Blank Page is an AI-powered app that uses a specially trained conversational agent and outstanding UX to help users find the words to more easily express themselves.
- Adaptable to a broad range of contexts, it is designed to deepen users' understanding of themselves, others, or their situation strengthening communication skills and improving overall mental health.



Why Mental Health Tools Like Blank Page™ Matter

- 1 in 3 college students experience anxiety or depression (Source: ACHA, 2023)
- Its approach is aligned with Professor James Pennebaker's research into the therapeutic value of expressive writing.
- Writing is therapeutic, but many struggle to start or stay consistent
- Blank Page™ offers a safe, AI-guided space to express and reflect
- Its aim is not to replace a therapist or AI to do the writing for user but rather to ask the right questions and provide the proper encouragement to enable the user to find their own words to say what they think or feel.



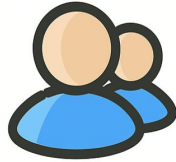
A five-step user journey from CA setup to reflection designed to build trust and emotional clarity

Blank Page features five steps:



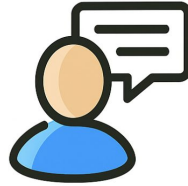
Personalization of CA

Users design a CA that feels relatable to build trust and comfort



Character Creation

Build profiles of real or fictional people to deepen emotional exploration



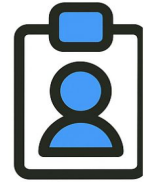
Conceiving

Plan expressive writing through structured dialogue with the CA



Writing & Revising

Draft and refine personal narratives using CA-guided prompts

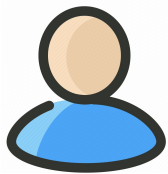


Reflection

Review writing to uncover insights and promote self-awareness



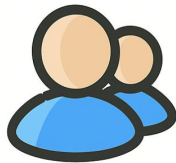
Centers on students, with therapist support and school integration to scale impact



Primary User

College & High Students

- High stress, transition periods, academic & social pressure
- Safe space to express emotions, consistent journaling support, trust without stigma



Secondary User

Therapists & Mental Health Caretakers

- Help users process and interpret emotions surfaced via journaling
- as a therapeutic supplement
- Offer structured feedback (with consent)



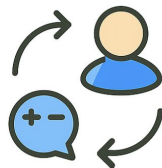
Institution Partnerships

University & School Collaborations

- campus wellness programs
- Integrate with student portals for seamless onboarding
- Enable anonymous usage to encourage adoption



Why Personalization of CA Emerged as Our MVP



Initial Customization

First interaction: user customizes the CA (name, demographic, tone)



Safe Expression Space

Enables safe, judgment-free environment to express difficult emotion



Interaction between the user and CA

the interaction between user and CA is representative of the interaction throughout the entire UX.



Especially Valuable for Students:

Especially impactful for high-school/college students who hesitate to open up.



Enabled real-time CA interaction with memory of user-defined traits like name and tone to foster emotional safety

The scope of the project involved:

1. Creating a functional prototype of the conversational agent element that reflects user-defined traits like name, background, and tone to build emotional safety and trust.

2. Enabled real-time interaction where the CA recalls user-defined attributes.



We compared fine-tuning and RLHF both offer empathy and nuance, but are costly, slow, and hard to iterate at MVP stage

Approach	Overview	Pros	Cons
Fine- tuning a language Model	Train a base model on therapy/counseling conversations	● Custom tone ● Tailored responses	● High compute + infra cost ● Time-intensive ● Risk of dataset bias ● costly & slow
Reinforcement learning with Human Feedback	Optimize LLM behavior using reward signals from human-labeled data	● Produces nuanced behavior ● Aligns with empathy goals	● Complex & expensive ● Needs large-scale human input ● Hard to iterate quickly



Used prompt engineering and memory layers to move fast, build trust, and scale effectively



Fastest Time-to-Market

- No model fine-tuning or RL pipeline needed we iterated quickly using OpenAI's GPT-4.1 via prompt engineering
- Enabled early user testing and feedback, critical for a startup validating product-market fit



Emotionally Aware Personalization

- Supabase memory layer stores emotional context, enabling nuanced, human-like continuity
- Builds emotional trust and long-term engagement essential in the mental wellness space



Business Fit

- Prioritizes engagement, scalability, and differentiation over technical novelty
- Supports a unique value proposition: expressive writing + empathetic dialogue, without drifting into regulated mental health territory



Built with React, FastAPI, GPT-4, and Supabase to enable fast iteration and memory recall, per sponsor direction

```
[React UI]
┆ API (FastAPI)
┆   LangChain + GPT-4
┆     Memory from Supabase
```

Core System Components

Front end

Using React (with likely Tailwind for styling and Vite as the dev server).

Back end

FastAPI handles the API, Supabase handles user/agent/message data.

AI engine

You're using langchain-openai to call GPT-4 with custom prompts.

Memory & Context

You're storing chat summaries + nicknames in Supabase and retrieving them for prompt context.



Simple login flows into a personalized AI journaling experience

Blank Page™[How It Works](#)[Benefits](#)[About](#)[Get Early Access](#)

Unlock Your Voice Through AI-Powered Writing

Blank Page helps you articulate thoughts and feelings through expressive writing, leading to deeper self-understanding and improved mental health.

[Start Your Journey →](#)[Watch Demo](#)

Your 5-Step Journey to Self-Discovery

Our AI-powered conversational agent guides you through a personalized writing experience designed to unlock deeper understanding of yourself and others.

[← Back to Home](#)

Welcome to Blank Page

Start your journey to better self-expression

[Sign In](#)[Sign Up](#)

Email

Password

[Sign In](#)

Users land on a welcoming screen encouraging them to begin their self-expression journey through AI-powered writing.

Users sign in or register via a minimalist login interface, unlocking access to personalized CA features and saved journaling progress.

A red location pin icon with a white circle in the center, pointing downwards. It is positioned in the bottom right corner of the slide, overlapping a blue and red geometric shape.

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Users create and manage CAs with traits that match their emotional needs

Your Dashboard

Manage your conversational agents

Sign Out

+ Create New Agent

 **angry old man**

sarcastic, rude

I am a dictator

Created 27/05/2025

D

 **Sarcastic**

sarcastic, rude

Sarcastic

Created 27/05/2025

D

 **Sia2**

calm

I am a fun loving introverted girlie pop

Created 27/05/2025

D

 **Sia**

AI Assistant

No background provided

Created 27/05/2025

D

 **try**

AI Assistant

No background provided

Created 26/05/2025

D

 **Hi**

AI Assistant

No background provided

Created 26/05/2025

D

 **New**

AI Assistant

No background provided

Created 26/05/2025

D

 **Friend**

AI Assistant

No background provided

Created 26/05/2025

D

 **Second Agent**

loving

Teacher

Created 26/05/2025

D

 **First Agent**

empathetic

Complete Agent Profile

Background

Agent Background

Personality Traits (comma-separated)

e.g. calm, empathetic

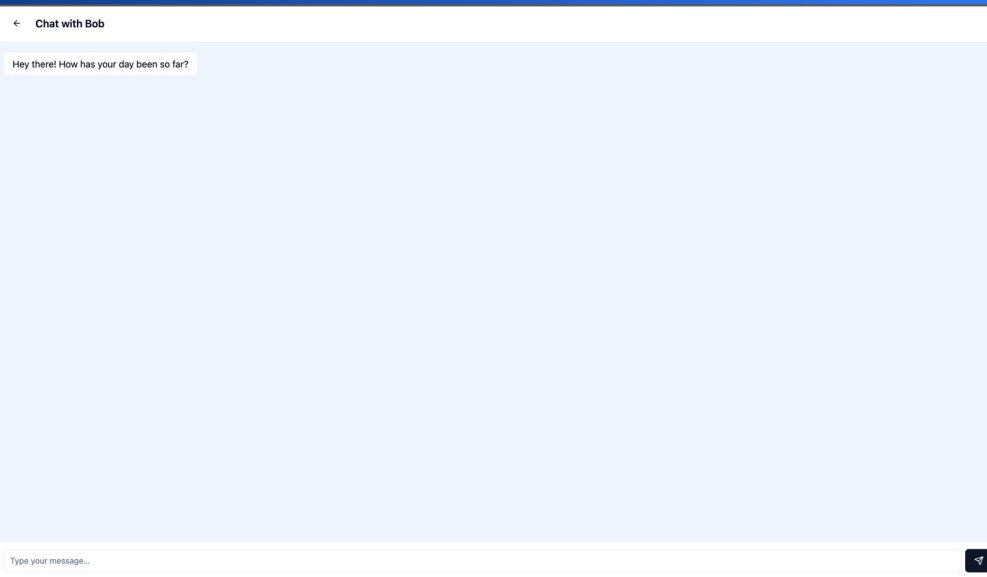
Save Profile

Users land on a centralized dashboard to view, manage, or create multiple conversational agents (CAs) tailored to their emotional needs.

Users can create, customize new CA by entering background and personality traits, e.g., calm, empathetic, sarcastic



Supports custom tone, memory of past chats, and context-aware responses for deeper engagement



Features of our MVP

1. Different tones and personality
2. CA is able to store their previous conversations in memory
3. CA is able to recall chats anecdotes from it's previous chat

Users engage in a real-time, open-ended chat with their customized CA (e.g., “Bob”), designed to initiate reflective, non-judgmental conversation.



Watch how our personalized CA guides reflective writing through memory, tone, and real-time interaction

MVP Demo



Thank you

